

## Vehicle Fires

### Arrival on Scene

- A. The Driver Operator will position the apparatus in a way that will provide the best protection for the crew during fire suppression activities
- B. The driver operator will engage the pump and stand by the pump panel for further instructions
- C. The Officer will provide a scene size-up, assume command, and develop a fire suppression plan

### Scene Safety

- A. Ensure that unauthorized/untrained personnel do not enter the hazardous area
- B. See HAZMAT procedures for vehicle fires involving Hazardous Materials
- C. Traffic Hazards

### Incident Actions

- A. The attack team will pull the hose line instructed by the Officer
- B. Additional firefighters will do as instructed by the officer
- C. The driver/operator will charge the hose line when directed and continue to monitor the pump operations
- D. The attack team will approach the vehicle slowly from the side extinguishing the fire as they approach. Caution should be taken as the team approached for exploding bumpers, fuel tanks, tires, etc.
- E. When the fire is under control the engine compartment and trunk of the vehicle on fire will be opened and cooled. The vehicles battery cables should be disconnected or cut when possible.

## Vehicle Accidents

### Arrival on Scene

- A. The Driver Operator will position the apparatus in a way that will provide the best protection for the crew during rescue and operation activities
- B. The Officer will provide a scene size-up, assume command, determine type of incident, number of vehicles involved, extent of damage to vehicles involved, extent of injuries, and if extrication is needed
- C. The Officer will request additional resources as needed (EMS, MedFlight)

### Scene Safety

- A. Ensure that unauthorized/untrained personnel do not enter the hazardous area
- B. Provide High Visibility PPE and other safety measures for Traffic Hazards

### Incident Actions

Command will assign teams as needed for:

- Fire and Hazard Control
- Extrication
- Patient Care
- Landing Zone
- Traffic Control

## Wildland Fires

### Arrival on Scene

- A. The Officer will provide a scene size-up, assume command, and develop a fire suppression plan
- B. The Officer will determine the location and size of the fire, direction and characteristics of fire travel, the fuel burning, and exposures
- C. The Officer will request additional resources as needed
- D. The Driver/Operator will park the apparatus in a safe, accessible location pointing away from the fire with the windows closed and the keys in the ignition
- E. Consider evacuations of citizens

### Scene Safety

- A. All personnel should know the location and direction of the fire travel
- B. Escape Plans shall be known to all fire personnel
- C. Be cautious for Spot fires
- D. Be cautious for Flare-ups
- E. Be aware of wind direction and velocity
- F. Be aware of topography
- G. Monitor crews for exhaustion
- H. Be aware of down wires, electric fences, etc.
- I. Be aware of equipment and personnel working above or around teams

### Incident Actions

- A. Base all actions and strategy on current and expected fire behavior
- B. Establish staging area for additional arriving apparatus and personnel
- C. Life safety and structural protection take priority over extinguishment of forest, brush, or ground cover
- D. If offensive attack is indicated, the head of the fire is to be attacked first. If that is not possible, the flanks should be attacked while working toward the head of the fire.
- E. If the fire is large and fast moving, then a direct attack may not be possible. In such cases, an indirect and/or parallel attack may be utilized by creating a fire line a distance ahead of the fire to halt the progress of the fire.
- F. Different methods of attack may be used simultaneously according to the situation
- G. Teams assigned to structural protection must keep hose lines flexible enough to be able to quickly break away in the event of being over run
- H. Collaborate with the DNR, \_\_\_\_\_ County Officials, and specialists/technicians as needed
- I. Communications and accountability of all incident personnel shall be maintained at all times.

## Hazardous Materials

**The Fire Department functions at the AWARENESS level.** At the operations level, we possess the basic knowledge necessary to protect the public from harm due to the exposure of hazardous materials. Prior to responding to a known or suspected hazardous materials incident, all personnel shall have Hazardous Materials training at the awareness level.

An incident involving hazardous materials can pose significant risk to the public and responding personnel. While all effort shall be made to protect the public, responders must NOT rush into the incident. Many HAZMAT incidents require the assistance, involvement, and response of other agencies; as such, mitigating the incident can take an extended period of time. All actions shall be based on informed decisions from qualified individuals.

Examples of activities and functions appropriate at the **AWARENESS** level include:

- Recognition of a Hazmat incident
- Recognition of hazards
- Identifying resource needs
- Initiating response of operations and/or technician level personnel
- Establishing scene control and management

### Arrival on Scene

- A. The Emergency Response Guidebook (ERG) shall be used to identify containers, chemicals involved, staging, and isolation zones.
- B. All units shall stage a safe distance away from the incident in accordance to ERG recommendations and current conditions. When selecting a staging area take into consideration the following: Wind direction, topography, accessibility, proximity to the incident, overhead obstacles, and potential for fire.
- C. The first arriving officer should establish command and complete an initial size-up
- D. Isolate the incident from all directions and evacuate as needed. Refuse admittance to the area. Note: In some situations, sheltering-in-place may be the most viable option.
- E. All persons who have been exposed to the material(s) shall be moved to a location where they are isolated from others and the incident so that they may be monitored and decontaminated if necessary.
- F. Obtain Safety Data Sheets and any chemical information available. This may include a, waybill, manifest, or other form of shipping papers if the incident involves the transportation of hazardous materials.
- G. Make contact with any company representatives and notify the appropriate agencies including a Wisconsin Level I HAZMAT Technician team if necessary.
- H. Assign an incident safety officer immediately. It is also suggested that the incident commander assign an assistant or scribe to begin documentation of all incident plans and operations.

### Scene Safety

- A. Establish hot, warm, and cold zones (utilize barrier tape and natural boundaries)
- B. Ensure that unauthorized/untrained personnel do not enter the hazardous area

### Incident Actions

- A. All incident actions shall not exceed the scope of the hazardous materials operations level which is limited to establishing command, identifying the materials involved, isolating the incident, and securing the scene. Other actions may occur only if the personnel have the appropriate level of PPE and training to SAFELY carry out the tasks associated with his/her assignments.
- B. An Incident Action Plan shall be created that identifies exposures, hazards, and incident objectives.

## Fire Department SOG's

- C. Dike and/or dam areas to contain run-off and prevent further contamination of other areas and water sources.
- D. Remove all ignition sources if materials are combustible or flammable in nature.
- E. Collaborate with the DNR, EPA, County Officials, company representatives, specialists/technicians, and the HAZMAT team as needed.
- F. Communications and accountability of all incident personnel shall be maintained at all times.

### **Reports and Documentation**

- A. The release of information regarding the incident shall be controlled by the OIC and the designated Public Information Officer. Only the OIC, County Emergency Management, or other designated Public Information Officer shall have the authority to provide public information.
- B. The Incident Action Plan and all operations performed on the incident shall be documented. A written report shall be completed by the OIC and filed with all other documents created.
- C. A log of all department personnel who were exposed or potentially exposed shall be maintained during the incident and filed along with other incident reports.
- D. A record of all other agencies and their corresponding personnel who responded shall also be kept.
- E. A record of all items used that will need to be replaced and any other expenses shall be kept during the incident and filed with the other incident reports.

### **Clean-Up**

Clean up of materials is the sole responsibility of the person or company responsible for the hazardous material(s) incident. All of the personnel and equipment in contact with the hazardous material(s) shall be decontaminated prior to returning to service.

## Natural Gas Incidents

Natural gas (Methane) is extremely flammable, colorless, odorless, and lighter than air. Mercaptan, with an odor of rotten eggs, is added to natural gas to help indicate its presence. The flammable range of natural gas is 4% Lower Explosive Limit (LEL) to 15% Upper Explosive Limit (UEL) by volume. Natural gas can displace oxygen and cause asphyxiation in certain settings, particularly confined spaces. The presence of natural gas in its flammable range should be evaluated with the use of a Combustible Gas Indicator (CGI) (on the Engine, Utility Company, or HAZMAT response unit.)

Fires involving natural gas should be controlled by stopping the flow of gas. In most cases, burning natural gas should not be extinguished as this would change the situation from a visible to an invisible hazard with an explosive potential. Natural gas is much lighter than air and will usually dissipate rapidly in the outside environment. Inside buildings, however, it tends to pocket, particularly in attics and dead space. Because natural gas is lighter than air, if confined it has the potential for a catastrophic explosion. Natural gas leaks above ground are much easier to manage than below ground leaks.

### Arrival on Scene

- A. Perform a scene size up and establish Command.
- B. The Emergency Response Guidebook shall be used to identify staging and isolation zones.
- C. Consciously avoid committing apparatus or personnel to a dangerous situation or a situation that could become dangerous due to gas migration.
- D. Attempt to locate the homeowner or responsible party and begin to gather information as to the hazard, potential victims, etc.
- E. Notify appropriate Utility/Gas Company and HAZMAT(if needed) for assistance and response
- F. Evacuate the area, working towards an "All Clear" of the immediate area and surrounding structures if necessary.
- G. Isolate the area/scene – Establishment of a "Hot Zone" is critical.
- H. Deny entry.
- I. If incident is determined to be a natural gas leak, ensure that personnel safety practices are in place when working in the Hot Zone including the wearing of the appropriate PPE with SCBA, face piece donned and breathing air. Establish a water supply and ensure that a charged hose line is in place. This line must be attended when Utility/Gas Company or HAZMAT personnel are securing the leak.
- J. Approach to the incident should be from upwind. Use wind speed/direction from dispatch and observe any on-scene indicators (trees, flags and any other items that may show direction).

### Scene Safety

- A. Establish hot, warm, and cold zones (utilize barrier tape and natural boundaries)
- B. Ensure that unauthorized/untrained personnel do not enter the hazardous area

### Incidents with an Explosion- Incident Actions

Units arriving on scene of an explosion must consider natural gas as a possible cause. Underground leaks may allow gas to travel considerable distances before entering a structure through the foundation, around pipes or through void spaces. Until it can be determined that the area is safe from danger of further explosions:

- A. Evacuate all civilians
- B. Keep the number personnel in the area to the minimum number necessary to stabilize the situation
- C. Don't rely on gas odor- Use a CGI to check all suspected areas

## Fire Department SOG's

- D. Check areas systematically using a CGI- Start outside of the area of the explosion and move into the area until readings indicate a detectable concentration.
- E. Map the readings for the affected area and relay it to Command
- F. If the gas concentration is encountered inside, adjacent to, or underneath any building, secure all possible sources of ignition in the affected area.
- G. Cut electricity from outside the affected area to avoid arcing.
- H. Ventilate buildings where gas is found with explosion-proof blowers only.
- I. Never enter inside when CGI is alarming for LEL.
- J. Ventilate first and wait for Utility/Gas Company or HAZMAT.
- K. Command shall provide for effective interaction between gas company personnel and the fire department.
- L. Command must ensure the safety and stability of the structure.

### **Incidents with No Explosion- Incident Actions**

Calls for "gas leak", gas odor", "broken gas line" and similar situations may range from minor to a potentially major incident. All these should be approached as potentially dangerous situations.

The following actions must occur:

- A. The first arriving unit with a CGI shall obtain a sufficient number of gas concentration readings for Command to evaluate the hazard and take appropriate action. (With these readings, a proper isolation perimeter can be established.)
- B. In all cases, personnel shall take whatever actions are necessary to provide for life safety and property conservation.
- C. Provide standby protection with a charged 1-3/4 inch hand line (fog nozzle) and a dry chemical extinguisher.
- D. Crews shall be in full protective equipment and SCBA.
- E. Crews should position themselves upwind.
- F. Evacuate any civilians in the area of escaping gas.
- G. Attempt to locate the source of the gas and any shutoff devices available to isolate and control the leak, i.e. water heater, stove, dryer, etc.

## Carbon Monoxide Incidents

Carbon monoxide (CO) is an odorless, tasteless, colorless gas that is deadly. It is a by-product of a fuel burning process. Many appliances such as furnaces, kitchen stoves, hot water heaters, automobiles, and etc. can produce carbon monoxide. When a faulty or unusual condition exists, carbon monoxide may be vented into areas where people are present. Carbon monoxide poisoning may be difficult to diagnose, its symptoms are similar to the flu, which may include headache, nausea, fatigue and dizzy spells.

### Arrival on Scene

- A. Perform a scene size up and establish Command.
- B. Attempt to locate the homeowner or responsible party and begin to gather information as to the hazard, potential victims, etc.
- C. Evacuation and treatment of occupants are the first priority.

### Scene Safety

- A. Establish hot, warm, and cold zones (utilize barrier tape and natural boundaries if necessary)
- B. Ensure that unauthorized/untrained personnel do not enter the hazardous area

### Incident Actions

- A. The CO meter should be activated outside of the structure to allow for fresh air calibration.
- B. Personnel shall use the CO meter and perform an initial CO level survey. The meter shall be used to survey the premises and verify the presence of carbon monoxide.
- C. Personnel shall enter with protective clothing and self-contained breathing apparatus (SCBA)
- D. If, at any time during the survey phase, the meters indicate a concentration of CO in excess of 20 ppm, or Oxygen by volume of less than 19.5%, then the survey team shall actively go on air
- E. Personnel shall initiate a primary search or verify that all occupants are outside of the structure.
- F. If occupants are showing signs of CO exposure, have EMS personnel treat them per their protocols and transport.
- G. Personnel should determine if the residential CO detector is in the low battery mode or alarm mode.
  - 1. A reading of 9 PPM or less does not require the evacuation of the premises:
    - a. Recommend the occupant, check their CO detector per manufacturer recommendations.
    - b. Attempt to reset the detector.
    - c. Inform occupants that if detector activates again, call 911.
  - 2. Any reading above 9 PPM but less than 100 PPM shall be considered above normal, and potentially dangerous to the occupants:
    - a. Recommend that all occupants leave the premises and begin ventilation.
    - b. Determine the source of the CO and shut down the appliance.
    - c. Once the premises have been reduced to a safe level of CO, the premises may be occupied at the discretion of the occupant.
    - d. Attempt to reset the detector
    - e. Inform occupants that if the detector activates again, call 911.
    - f. Inform the occupants of the actions that have taken place and that they should notify the appropriate resources for assistance
  - 3. Emergency Operations Reading of 100 PPM or greater:
    - a. Order the occupants to leave the premises immediately.



## Fire Department SOG's

- b. Determine the source of the carbon monoxide and shut down the appliance.
- c. Call for the Utility/Gas Company to respond to the scene if necessary
- d. Ventilate the structure and reduce CO to a safe level.
- e. Once the carbon monoxide level is safe the structure may be occupied at the discretion of the occupant.
- f. Inform the occupants of the actions that have taken place and they should notify the appropriate resources for assistance

## Confined Space Rescue

### Definitions

Confined space: A confined space is any area or vessel, which meets all 3 of the following:

1. Is large enough and so configured that an employee can enter and perform work
2. Has limited means of entry or exit
3. Is not designed for continuous occupancy

Permit required confined space: A permit required confined space is defined as a confined space which has one or more of the following:

1. Contains or has a potential to contain a hazardous atmosphere
2. Contains a material with potential for engulfment
3. Is so structured that an entrant could become trapped or asphyxiated
4. Contains any other recognized serious safety or health hazard – i.e. moving parts, noise

Recovery mode: Recovery mode is defined as situations where the victim is obviously expired or after a period of time during the rescue operation where time, conditions, or other factors have reduced the chance for the victim's survival to minimal.

Rescue mode: Rescue mode is defined as situations where the victim is believed or known to be alive. If this is unknown, personnel should operate in the rescue mode until time, conditions, or other elements make the chance for survival minimal.

Confined space rescue operations present a significant danger to fire department personnel. The safe and effective management of these operations requires special considerations and resources. Examples of possible confined spaces includes tunnels, sewers, tanks, process vessels, manholes, storm drains, furnaces, silos, and industrial spaces.

**The Fire Department functions at the AWARENESS level. Therefore it is the policy of the Fire Department that personnel SHALL NOT enter into a confined space. Entry is considered to have occurred as soon as any part of an entrant's body breaks the plane of an opening into the space.**

Examples of activities and functions appropriate at the **AWARENESS** level include:

- Recognition of a confined space incident
- Recognition of confined space hazards
- Performing a **non-entry** retrieval
- Identifying resource needs
- Initiating response of operations and/or technician level personnel
- Establishing scene control and management

### Arrival On Scene

- A. The first-in unit should position the apparatus appropriately
- B. The first arriving officer should establish command and complete an initial size-up including:
  1. Secure any witnesses
  2. Obtain the confined space entry permit and any other available information
  3. Location, number, condition of victims, and length of time in confined space
  4. Utility and other scene hazards – i.e. hazardous materials, low oxygen levels
  5. Type of work being performed in the confined space
  6. Type of PPE being used by victim(s)
  7. Determination of rescue or recovery mode
  8. Determination of additional resources needed

## \_\_\_\_\_ Fire Department SOG's

### **Scene Safety**

- A. Establish hot, warm, and cold zones (utilize barrier tape and natural boundaries)
- B. Ensure that unauthorized/untrained personnel do not enter the confined space
- C. Confirm or implement lock out/tag out

### **Incident Actions**

- A. If victim is attached to a body harness and retrieval line, the rescuers may lift the victim from the confined space area
- B. Attempt to establish contact with victim(s)
- C. Establish atmospheric monitoring
- D. Establish ventilation of confined space after atmospheric monitoring
- E. If safe to do so and if it can be accomplished from outside the confined space, shutdown non-essential equipment that is located within the confined space
- F. Establish staging area for additional arriving apparatus and personnel

## Trench Rescue

### Definitions

Recovery mode: Recovery mode is defined as situations where the victim is obviously expired or after a period of time during the rescue operation where time, conditions, or other factors have reduced the chance for the victim's survival to minimal.

Rescue mode: Rescue mode is defined as situations where the victim is believed or known to be alive. If this is unknown, personnel should operate in the rescue mode until time, conditions, or other elements make the chance for survival minimal.

Trench: An excavation in which the depth is greater than the width and is less than 15 feet wide.

Trench rescue operations present a significant danger to fire department personnel. The safe and effective management of these operations requires special considerations and resources.

**The Fire Department functions at the AWARENESS level. Therefore it is the policy of the Fire Department that personnel SHALL NOT enter into an unsafe trench or excavation.**

Examples of activities and functions appropriate at this level include:

- Recognition of a trench collapse incident
- Identifying resource needs
- Initiating response of operations and/or technician level personnel
- Establishing scene control

### Arrival on Scene

- A. The first-in unit should position the apparatus a minimum of 50' from the location of the trench collapse. Additional arriving units should initially stage a minimum of 150' from the location.
- B. The first arriving officer should establish command and complete an initial size-up including:
  1. Secure any witnesses
  2. Location, number, condition of victims and how long buried
  3. Depth of trench
  4. Utility and other scene hazards
  5. Determination of rescue or recovery mode
  6. Determination of additional resources needed

### Scene safety

- A. Establish hot, warm, and cold zones (utilize barrier tape and natural boundaries)  
Hot – 0-100' from trench      Warm – 100-500' from trench      Cold – 500' and further from trench
- B. Secure and/or shut down machinery and traffic within 300' of trench
- C. Implement lock out/tag out
- D. Place ground pads within 4 feet of trench

### Incident Actions

- A. If victim is partially buried, lower lifeline and instruct victim to tie around themselves
- B. If indicated, lower ladder into trench - for victim self-rescue only
- C. If victim is buried, mark last known location using dry chemical extinguisher
- D. Establish atmospheric monitoring
- E. Establish ventilation of trench (if necessary)
- F. Relocate any soil piles to be a minimum of 2 feet from trench
- G. Establish staging area for additional arriving apparatus and personnel

## Water Rescue

### Arrival on Scene

- A. The Officer will provide a scene size-up, assume command, develop a plan, and call for additional technical rescue mutual aid units as needed.
- B. The Officer will secure a responsible party or witness.
- C. The Officer will determine if the operation will be a "Rescue" or "Recovery."
- D. The Officer will ensure that a communication system has been coordinated for all activities.
- E. The Rescue Team (Consisting of members trained in water rescue operations) will:
  - a. Assume rescue/recovery operations control.
  - b. Identify hazards and critical factors.
  - c. Develop a plan and back-up plan.
  - d. Communicate with resources.
  - e. Inform command of conditions, actions and needs during all phases of the operation.

### Scene Safety

- A. Make the General Area Safe.
  - a. Establish a Hazard Zone Perimeter.
  - b. Keep all non-essential personnel out of the hazard zone.
  - c. Remove all non-essential civilians from the hazard zone.
- B. A Designated Safety Officer must be assigned for the scene.
  - a. Identify hazards present and if possible, have them secured.
  - b. Notify personnel of hazards (volume, velocity, water temperature, floating debris, unusual drop offs, hydraulic effects, depth of water, inclement weather, etc.)
- C. Assign Personnel Up Stream that can notify Command and the Rescue team of any upstream hazards that may affect the operation.
- D. Assign Personnel Down Stream that can throw ropes or watch for rescue personnel or victims that maybe washed downstream. \*Personnel working near the water's edge must wear appropriate PPE/PFD
- E. Assemble all necessary personnel, equipment, and patient packaging equipment that will be required for the rescue operation according to the action plan.

### Incident Actions

- A. The Rescue Plan. Rescue plans should be conducted from low risk to high risk order.
  - a. **TALK-** If the water is calm or slow moving, try to talk the victim into self-rescue if possible
  - b. **REACH-** Extend an arm, pike pole, rescue hook, or any other such object to reach the victim and pull them from the water.
  - c. **THROW-** Attempt to throw the victim a throw-bag rescue line or some other type of approved safety flotation device and "pendulum-belay" or "haul" the victim to the bank.
  - d. **ROW-** If it is determined that a boat-based operation shall be utilized, Mutual Aid must be called. Command and the Mutual Aid responding Technical Rescue Team will establish an action plan involving the use of boat-based operations, anchors, and rope systems.
  - e. **GO-** If it is not possible to ROW to the victim, although very high risk, putting a rescuer(s) in the water can be considered. **Only rescuers with proper water rescue training, PPE/PFD, and equipment may enter the water.** Prior to entering the water, the rescuer(s) must discuss the action plan, including specific tasks and objectives, hazards, emergency procedures, and alternate plans. The rescuer(s) shall never be attached to a life line without the benefit of a quick release mechanism. The rescuer(s) should take at the least a PFD to the victim.

## Fire Department SOG's

The rescuer(s) shall not do a breath-hold surface dive in an attempt to locate a victim beneath the surface of the water.

f. **HELO-** Helicopter operations are considered high-risk and shall be decided upon proper consultation with local law enforcement, dispatch, and appropriate authorities. Command must also determine that a rescue-qualified pilot is available for the rescue operation. If so, the Pilot will have the final say on, if and how, the helicopter will be used in the rescue operation.

B. Assess the Victim

a. When the rescuer(s) reach the victim, a primary survey shall be completed.

Assess Airway, Breathing, Circulation, and exact method of entrapment.

b. If the victim is conscious, rescuer(s) should determine if the victim can assist in the rescue.

c. If the victim is unconscious, the rescue must be completed as quickly as possible.

d. If it has been determined that the operation has become an underwater "recovery," the operation shall proceed as a dive operation. A Mutual Aid Technical Rescue Dive Team shall be called to perform the dive operation.

C. Treatment

a. Assist the victim to safety on shore and turn over to EMS/ALS for immediate assessment.

b. Assist with initiating C-Spine precautions as soon as possible.

c. Assist with treating any life threatening conditions.

**Termination**

A. Ensure personnel accountability.

B. Consider decontamination of victim(s) and rescuer(s.)

C. Recover all tools and equipment used in the rescue/recovery. In the cases of a fatality, consider leaving everything in place until the investigative process has concluded.